

```
> restart;
```

```
> with(linalg):
```

Warning, the protected names norm and trace have been redefined and unprotected

```
> V:=array(1..3, [rho, u, p]);
```

$$V := \begin{bmatrix} \rho & u & p \end{bmatrix} \quad (1)$$

```
> U:=array(1..3, [rho, rho*u, rho*u^2/(2)+p/(g-1)]);
```

$$U := \begin{bmatrix} \rho & \rho u & \frac{1}{2} \rho u^2 + \frac{p}{g-1} \end{bmatrix} \quad (2)$$

```
> F:=array(1..3, [rho*u, rho*u^2+p, rho*u^3/(2)+(g/(g-1))*u*p]);
```

$$F := \begin{bmatrix} \rho u & \rho u^2 + p & \frac{1}{2} \rho u^3 + \frac{g u p}{g-1} \end{bmatrix} \quad (3)$$

```
> dFdV:=array(1..3, 1..3);
```

$$dFdV := \text{array}(1..3, 1..3, [\]) \quad (4)$$

```
> dFdV:=jacobian(F, V);
```

$$dFdV := \begin{bmatrix} u & \rho & 0 \\ u^2 & 2 \rho u & 1 \\ \frac{1}{2} u^3 & \frac{3}{2} \rho u^2 + \frac{g p}{g-1} & \frac{g u}{g-1} \end{bmatrix} \quad (5)$$

```
> dUdV:=array(1..3, 1..3);
```

$$dUdV := \text{array}(1..3, 1..3, [\]) \quad (6)$$

```
> dUdV:=jacobian(U, V);
```

$$dUdV := \begin{bmatrix} 1 & 0 & 0 \\ u & \rho & 0 \\ \frac{1}{2} u^2 & \rho u & \frac{1}{g-1} \end{bmatrix} \quad (7)$$

```
> dVdU:=array(1..3, 1..3);
```

$$dVdU := \text{array}(1..3, 1..3, [\]) \quad (8)$$

```
> dVdU:=inverse(dUdV);
```

$$dVdU := \begin{bmatrix} 1 & 0 & 0 \\ -\frac{u}{\rho} & \frac{1}{\rho} & 0 \\ \frac{1}{2} u^2 (g-1) & -u (g-1) & g-1 \end{bmatrix} \quad (9)$$

```
> dFdU:=array(1..3, 1..3);
      dFdU:= array(1..3, 1..3, [ ]) (10)
```

```
> dFdU:=evalm(dFdV&*dVdU):
> for i from 1 to 3 by 1 do for j from 1 to 3 by 1 do dFdU[i,j]:=
  factor(simplify(dFdU[i,j])) od: od:
> evalm(dFdU);
```

```
[[ [ 0, 1, 0 ], (11)
```

$$\begin{bmatrix} \frac{1}{2} u^2 (-3 + g), -u (-3 + g), g - 1 \\ \frac{1}{2} \frac{u (-3 \rho u^2 g + 2 \rho u^2 - 2 g p + \rho u^2 g^2)}{(g - 1) \rho}, \\ -\frac{1}{2} \frac{-5 \rho u^2 g + 3 \rho u^2 - 2 g p + 2 \rho u^2 g^2}{(g - 1) \rho}, g u \end{bmatrix}$$

```
> A:=array(1..3,1..3,[[u, rho, 0], [0, u, 1/rho], [0, g*p, u]]);
```

$$A := \begin{bmatrix} u & \rho & 0 \\ 0 & u & \frac{1}{\rho} \\ 0 & g p & u \end{bmatrix} \quad (12)$$

```
> assume(g, real);
> assume(g > 0);
> assume(rho, real);
> assume(rho > 0);
> assume(a, real);
> assume(a > 0);
> assume(p, real);
> p:=rho*a^2/g;
```

$$p := \frac{\rho \sim a \sim^2}{g \sim} \quad (13)$$

```
> I3:=array(1..3, 1..3, identity);
      I3:= array(identity, 1..3, 1..3, [ ]) (14)
```

```
> c1:=factor(simplify(det(A-lambda*I3)));
      c1 := (-u + λ) (λ - u + a~) (-λ + u + a~) (15)
```

```
> R1:=solve(c1=0, lambda);
      R1 := u, u + a~, u - a~ (16)
```

```
> c2:=factor(simplify(det(dFdU-lambda*I3)));
```

$$c2 := (-u + \lambda) (\lambda - u + a\sim) (-\lambda + u + a\sim) \quad (17)$$

```
> R2:=solve(c2=0, lambda);
```

$$R2 := u, u + a\sim, u - a\sim \quad (18)$$

```
> CC:=array(1..3,1..3):
```

```
> CC:=evalm(A-evalm((u-a)*I3)):
```

```
> for i from 1 to 3 by 1 do for j from 1 to 3 by 1 do CC[i,j]:=
simplify(subs(p=rho*a^2/g, CC[i,j])) od: od:
```

```
> nullspace(CC);
```

$$\left\{ \left[\begin{array}{ccc} -\frac{\rho\sim}{a\sim} & 1 & -\rho\sim a\sim \end{array} \right] \right\} \quad (19)$$

```
> CC:=evalm(A-evalm(u*I3)):
```

```
> for i from 1 to 3 by 1 do for j from 1 to 3 by 1 do CC[i,j]:=
simplify(subs(p=rho*a^2/g, CC[i,j])) od: od:
```

```
> nullspace(CC);
```

$$\left\{ \left[\begin{array}{ccc} 1 & 0 & 0 \end{array} \right] \right\} \quad (20)$$

```
> CC:=evalm(A-evalm((u+a)*I3)):
```

```
> for i from 1 to 3 by 1 do for j from 1 to 3 by 1 do CC[i,j]:=
simplify(subs(p=rho*a^2/g, CC[i,j])) od: od:
```

```
> nullspace(CC);
```

$$\left\{ \left[\begin{array}{ccc} 1 & \frac{a\sim}{\rho\sim} & a\sim^2 \end{array} \right] \right\} \quad (21)$$

```
> CC:=evalm(dFdU-evalm((u-a)*I3)):
```

```
> for i from 1 to 3 by 1 do for j from 1 to 3 by 1 do CC[i,j]:=
simplify(subs(p=rho*a^2/g, CC[i,j])) od: od:
```

```
> nullspace(CC);
```

$$\left\{ \left[\begin{array}{ccc} 1 & u - a\sim & -\frac{1}{2} \frac{u^2 + 2 g\sim u a\sim - 2 a\sim^2 - u^2 g\sim - 2 u a\sim}{g\sim - 1} \end{array} \right] \right\} \quad (22)$$

```
> CC:=evalm(dFdU-evalm(u*I3)):
```

```
> for i from 1 to 3 by 1 do for j from 1 to 3 by 1 do CC[i,j]:=
simplify(subs(p=rho*a^2/g, CC[i,j])) od: od:
```

```
> nullspace(CC);
```

$$\left\{ \left[\begin{array}{ccc} 1 & u & \frac{1}{2} u^2 \end{array} \right] \right\} \quad (23)$$

```
> CC:=evalm(dFdU-evalm((u-a)*I3)):
```

```
> for i from 1 to 3 by 1 do for j from 1 to 3 by 1 do CC[i,j]:=
simplify(subs(p=rho*a^2/g, CC[i,j])) od: od:
```

```
> nullspace(CC);
```

$$\left\{ \left[\begin{array}{ccc} 1 & u - a \sim & -\frac{1}{2} \frac{u^2 + 2 g \sim u a \sim - 2 a \sim^2 - u^2 g \sim - 2 u a \sim}{g \sim - 1} \end{array} \right] \right\} \quad (24)$$

```
> lambda:=array(1..3):
```

```
> lambda[1]:=u-a;
```

$$\lambda_1 := u - a \sim \quad (25)$$

```
> lambda[2]:=u;
```

$$\lambda_2 := u \quad (26)$$

```
> lambda[3]:=u+a;
```

$$\lambda_3 := u + a \sim \quad (27)$$

```
> r1:= vector(3, [1, -a/rho, a^2]);
```

$$r1 := \left[\begin{array}{ccc} 1 & -\frac{a \sim}{\rho \sim} & a \sim^2 \end{array} \right] \quad (28)$$

```
> r2:= vector(3, [1, 0, 0]);
```

$$r2 := \left[\begin{array}{ccc} 1 & 0 & 0 \end{array} \right] \quad (29)$$

```
> r3:= vector(3, [1, a/rho, a^2]);
```

$$r3 := \left[\begin{array}{ccc} 1 & \frac{a \sim}{\rho \sim} & a \sim^2 \end{array} \right] \quad (30)$$

```
> res0:=vector(3);
```

$$res0 := array(1..3, []) \quad (31)$$

```
> res:=vector(3);
```

$$res := array(1..3, []) \quad (32)$$

```
> res0:=evalm((A&r1)-lambda[1]*r1):
```

```
> for i from 1 to 3 by 1 do res[i]:=factor(simplify(expand(res0[i]))
)) od:
```

```
> evalm(res);
```

$$\left[\begin{array}{ccc} 0 & 0 & 0 \end{array} \right] \quad (33)$$

```
> res0:=evalm((A&r2)-lambda[2]*r2):
```

```
> for i from 1 to 3 by 1 do res[i]:=factor(simplify(expand(res0[i]))
)) od:
```

```
> evalm(res);
```

$$\left[\begin{array}{ccc} 0 & 0 & 0 \end{array} \right] \quad (34)$$

```
> res0:=evalm((A&r3)-lambda[3]*r3):
```

```
> for i from 1 to 3 by 1 do res[i]:=factor(simplify(expand(res0[i]))
)) od:
```

```
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \quad (35)$$

```
> rc1 := vector(3):
> rc2 := vector(3):
> rc3 := vector(3):
> rtest:=vector(3):
> rtest:=evalm(dUdV&*r1):
> for i from 1 to 3 by 1 do rc1[i]:=simplify(rtest[i]) od:
> evalm(rc1);
```

$$\begin{bmatrix} 1 & u - a \sim & -\frac{1}{2} \frac{-u^2 g \sim + u^2 + 2 u a \sim g \sim - 2 u a \sim - 2 a \sim^2}{g \sim - 1} \end{bmatrix} \quad (36)$$

```
> rtest:=evalm(dUdV&*r2):
> for i from 1 to 3 by 1 do rc2[i]:=simplify(rtest[i]) od:
> evalm(rc2);
```

$$\begin{bmatrix} 1 & u & \frac{1}{2} u^2 \end{bmatrix} \quad (37)$$

```
> rtest:=evalm(dUdV&*r3):
> for i from 1 to 3 by 1 do rc3[i]:=simplify(rtest[i]) od:
> evalm(rc3);
```

$$\begin{bmatrix} 1 & u + a \sim & \frac{1}{2} \frac{u^2 g \sim - u^2 + 2 u a \sim g \sim - 2 u a \sim + 2 a \sim^2}{g \sim - 1} \end{bmatrix} \quad (38)$$

```
> R:=array(1..3,1..3, sparse):
> for i from 1 to 3 by 1 do R[i,1]:=simplify(r1[i]) od:
> for i from 1 to 3 by 1 do R[i,2]:=simplify(r2[i]) od:
> for i from 1 to 3 by 1 do R[i,3]:=simplify(r3[i]) od:
> L:=array(1..3,1..3, sparse):
> L:=inverse(R):
> l1 := vector(3):
> l2 := vector(3):
> l3 := vector(3):
> for i from 1 to 3 by 1 do l1[i]:=simplify(L[1,i]) od:
> for i from 1 to 3 by 1 do l2[i]:=simplify(L[2,i]) od:
> for i from 1 to 3 by 1 do l3[i]:=simplify(L[3,i]) od:
> evalm(l1);
```

$$\begin{bmatrix} 0 & -\frac{1}{2} \frac{\rho \sim}{a \sim} & \frac{1}{2} \frac{1}{a \sim^2} \end{bmatrix} \quad (39)$$

```
> evalm(l2);
```

